A PROJECT REPORT ON

OIL SLICK DETECTION AND DRIFT PREDICTION USING SAR IMAGERY

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CERTIFICATE

This is to certify that the project report entitled

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is a bonafide work carried out by them under the supervision of **Prof.M.R.Apsangi** and is approved for partial fulfillment of the requirement of Savitribai Phule Pune University for the award of the Degree of Bachelor of Engineering (Information Technology).

This project report has not been earlier submitted to any other Institute or University for the award of any degree or diploma.

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NOMENCLATURE

Sr. No.	Short Form	Full Form
1.	SAR	Synthetic Aperture Radar
2.	CNN	Convolutional Neural Network
3.	H	Horizontal
4.	V	Vertical
5.	HH	Horizontal Horizontal
6.	VV	Vertical Vertical
7.	HV	Horizontal Vertical
8.	VH	Vertical Horizontal

CONTENTS

ABSTRACT

This project presented an intelligent system for detecting and forecasting the movement of marine oil slicks using satellite-based Synthetic Aperture Radar (SAR) imagery. The aim was to automate the process of identifying oil spills and predicting their drift patterns under varying environmental conditions. SAR images were preprocessed to remove noise and enhance quality, followed by the application of deep learning models such as Convolutional Neural Networks (CNN) and ResNet for accurate detection and segmentation of oil slick regions. The system further integrated environmental parameters including wind speed, ocean currents, and temperature to predict the movement of detected slicks using hybrid drift models based on Lagrangian and Eulerian approaches. Experimental results showed that the proposed approach effectively reduced false detections and provided reliable drift forecasts, thereby improving response planning and environmental protection efforts. The developed framework demonstrated the potential for real-time, scalable marine monitoring and can be extended for broader ocean surveillance applications.

Keywords: Oil Slick Detection, Drift Prediction, Satellite Imagery, SAR, Deep Learning, Lagrangian Model, Environmental Monitoring.

CHAPTER 1

INTRODUCTION

Oil slicks-thin layers of petroleum floating on the ocean surface-have become a critical environmental concern due to their rapid spread and severe ecological consequences. While some oil reaches the surface through natural seepage from the seafloor, most slicks originate from human activities such as tanker accidents, pipeline failures, offshore drilling incidents, and improper disposal of ship waste. Even a small quantity of oil can expand over vast oceanic areas, blocking sunlight and oxygen exchange essential for microscopic phytoplankton, which form the foundation of the marine food chain. Marine organisms may ingest oil directly or through contaminated prey, while seabirds lose their waterproofing ability, often resulting in irreversible harm. These cascading effects highlight the urgent need for efficient detection and monitoring of oil spills to protect marine ecosystems.

Traditional approaches to oil spill monitoring relied heavily on ship patrols and aerial surveys. Although useful, these methods are expensive, resource-intensive, and severely limited in spatial coverage. Ships can only inspect narrow regions at a time, requiring large crews, high fuel consumption, and extensive planning. Aerial surveys, while faster, rely on skilled personnel, specialized aircraft, and favourable weather conditions. Most importantly, both methods observe only specific locations at specific times, making it impossible to maintain continuous watch over dynamic ocean environments. As oil slicks disperse rapidly due to winds and currents, delays in detection can significantly increase the environmental and economic losses associated with spill events.

The advent of satellite-based observation has transformed the field of marine oil spill monitoring by overcoming the limitations of conventional surveillance. Satellites provide large-scale, repeated coverage of the world’s oceans, including remote or inaccessible regions. They can operate under cloudy, stormy, or nighttime conditions, making them indispensable for early detection. Real-time or near-real-time data enables authorities to respond faster, issue public alerts, and coordinate cleanup operations more effectively. Continuous monitoring also allows experts to study the evolution and drift of slicks over long periods, improving the accuracy of forecasting and facilitating quick mitigation actions.

Table 1.1.1: Comparison of satellite sensors and their characteristics

Satellite	Sensor type	Spatial Resolution	Revisit time
Sentinel-1	C-Band SAR	~ 10 m	6–12 days
Sentinel-2	Multispectral Optical	10–60 m	5 days
RADARSAT	C-Band SAR	3–100 m	24 hrs
TerraSAR-X	X-band SAR	~1–3 m	Depends on orbit & tasking
RISAT-2	X-band SAR	~1 m	Depends on orbit & tasking
Cartosat-2	Panchromatic optical	~1–3 m	5 days

Among satellite instruments, Synthetic Aperture Radar (SAR) has become one of the most powerful tools in oil spill detection. SAR sensors such as those aboard the Sentinel-1 mission generate high-resolution images irrespective of weather or illumination conditions. Oil slicks appear as dark formations in most SAR imagery because they dampen the small surface waves responsible for radar backscatter. However, natural phenomena like biogenic films, low-wind areas, or organic debris can resemble oil spills, complicating detection. Early detection methods relied on simple thresholding or manual visual inspection, while later approaches adopted machine learning models like Support Vector Machines and Random Forests. Today, deep learning architectures including CNNs, U-Net, and SegNet dominate the field, achieving higher accuracy but still facing challenges such as false positives and limited availability of labeled training data.

Understanding where an oil slick will move after detection is equally crucial. Drift prediction models seek to estimate the future trajectory of oil patches by incorporating environmental variables such as winds, surface currents, wave fields, spreading behaviour, evaporation, and chemical breakdown of oil. Lagrangian models track individual oil particles, while Eulerian models estimate concentration changes over spatial grids. Integrating SAR imagery with hydrodynamic simulations and in-situ measurements from buoys or coastal sensors significantly enhances prediction reliability. This creates an end-to-end system in which satellites detect the spill, algorithms identify it, and drift models forecast its movement-allowing response teams to strategically deploy containment equipment.

A key component in refining detection accuracy is the use of different SAR polarization modes, particularly those employed by Sentinel-1. Polarization refers to the orientation of radar waves during transmission and reception either horizontal (H) or vertical (V). Single-polarization modes (VV or HH) capture co-polarized signals and are simple to process but provide limited surface information. Dual-polarization modes, such as VH or HV, transmit in one polarization and receive in both co- and cross-polarized channels. These richer datasets reveal detailed scattering properties of the sea surface, enabling superior classification performance. For oil spill detection, dual-polarization SAR generally enhances reliability by distinguishing true oil slicks from look-alike phenomena.

Table 1.1.2: Sentinel-1 Modes

Polarization	**Meaning**	**Modes**
VV	Transmit Vertical, Receive Ver- tical	Single Polarization
HH	Transmit Horizontal, Receive Horizontal	Single Polarization
HV	Transmit Horizontal, Receive Vertical	Dual Polarization (HH + HV)
VH	Transmit Vertical, Receive Hori- zontal	Dual Polarization (VV + VH)

Overall, the integration of advanced satellite technologies, improved SAR imaging capabilities, and robust computational models has significantly strengthened global oil spill monitoring and response efforts. Despite persistent challenges—such as look-alike surface features, rapidly changing environmental conditions, and the ongoing need for larger labeled datasets—the field continues to evolve. Future advancements may emerge from multi-sensor fusion (combining drones, satellites, and ground systems), hybrid human-AI decision frameworks, and next- generation deep learning models. As these technologies mature, they offer the promise of more accurate detection, faster response, and ultimately, better protection of vulnerable marine ecosystems.

1.1 AIM

To develop a system that detects oil slicks on satellite SAR imagery using deep learning and predicts their drift using environmental data, enabling faster and more accurate monitoring and decision-making for marine pollution response.

1.2 MOTIVATION

The motivation for this project arises from the urgent need to detect and manage oil spills quickly to reduce environmental damage. Oil slicks spread rapidly and can harm marine life, fisheries, and coastal ecosystems, making early identification critical. Traditional monitoring methods are slow, costly, and limited by weather and visibility, leading to delays in response. Using SAR satellite imagery combined with deep learning enables faster, more accurate detection of oil slicks without being affected by clouds, lighting, or distance. Additionally, predicting the drift of detected slicks helps authorities plan containment measures in advance, reducing the area affected by the spill. This project aims to create an automated system that improves monitoring efficiency, supports informed decision-making, and strengthens environmental protection efforts.

1.3 OBJECTIVE

- Apply image processing and deep learning to reduce false alarms
- Improving the prediction accuracy and generalizing the model
- Integrate environmental data (currents, wind, tides) for drift prediction
- Forecast slick movement and impacted areas through predictive modeling
- Develop a unified, scalable system for real-time monitoring and response

CHAPTER 2
LITERATURE SURVEY

Detection of oil spills in SAR satellite images using the Otsu-Bradley thresholding method [1] focuses on automatically identifying oil-contaminated regions on the ocean surface. The study combines two thresholding techniques: the global thresholding of Otsu and Bradley's local adaptive thresholding. The Otsu method separates potential slick areas from water based on the overall image intensity, while the Bradley method refines the segmentation by adapting to local brightness variations. This hybrid approach improves detection accuracy, reduces false positives caused by calm water or natural slicks, and does not require training data. The method is computationally efficient and suitable for fast real-time oil slick detection in remote sensing applications.

Textural Features for Image Classification [2] focuses on using texture as a measurable property for image analysis and classification. The study introduces the Gray-Level Co-occurrence Matrix (GLCM), which captures how often different pixel intensity pairs occur at specific spatial distances and directions in an image. From this matrix multiple statistical texture descriptors are derived, such as contrast, correlation, energy, and homogeneity.The paper demonstrates that these textural descriptors help differentiate image regions that may have similar intensity but different surface patterns. This makes texture-based classification more reliable in applications such as remote sensing, medical imaging, and pattern recognition. The contribution of this work established GLCM as one of the standard approaches to texture feature extraction and improved the accuracy and robustness of image classification tasks.

Oil Spill Detection using Convolutional Neural Networks [3] and Sentinel-1 SAR Imagery focuses on using deep learning to automatically identify oil slicks from satellite SAR images. Sentinel-1 provides radar images that capture the ocean surface in any weather or lighting conditions, making it suitable for real-time monitoring. A Convolutional Neural Network (CNN) is trained on labeled SAR image patches that contain oil spills and non-oil areas. The model learns to recognize patterns and textural differences between actual oil slicks and look-alike features such as calm water or natural films. By extracting spatial features directly from the imagery, the CNN eliminates the need for hand-crafted feature design. The study shows that CNN-based detection achieves higher accuracy and fewer false positives compared to traditional image-processing or thresholding methods. It also improves detection efficiency by reducing manual analysis time, making it useful for large-scale oil spill monitoring and emergency response.

Dark Spot Detection in SAR Images of Oil Spill Using SegNet [4] focuses on detecting oil slicks (dark spots) in SAR satellite imagery using a deep learning segmentation model. The study applies the SegNet architecture, which is an encoder-decoder CNN designed for pixel-wise image segmentation. The encoder extracts spatial and textural features from the SAR image, while the decoder reconstructs a segmented map where dark regions corresponding to oil slicks are highlighted. This automated segmentation reduces the need for manual inspection and minimizes false detections caused by look-alike dark patterns such as calm water or natural slicks. The method improves accuracy and consistency by learning complex spatial characteristics of oil slicks directly from data. It is computationally efficient and suitable for large-scale satellite image processing, making it valuable for oil spill monitoring and rapid response systems.

U-Net: Convolutional Networks for Biomedical Image Segmentation [5] introduces a deep learning model built specifically for segmentation tasks in medical images. The architecture is based on convolutional neural networks and is designed to improve learning efficiency while maintaining high accuracy, even with limited training data. The network follows an encoder-decoder structure. The encoder extracts features from the input image, while the decoder reconstructs a segmented output. Skip connections between corresponding layers help retain spatial details that are often lost during feature extraction. This makes the segmentation more precise. The model is computationally efficient and produces accurate pixel-level segmentation results. It became widely used in biomedical applications because it reduces training complexity and performs well on tasks such as identifying organs, tissues, or abnormalities in medical scans.

OpenDrift v1.0: A Generic Framework for Trajectory Modelling [6] introduces an open-source and flexible system used for predicting how objects or particles move in the ocean or atmosphere over time. The framework uses real environmental data such as wind, ocean currents, and wave conditions to simulate the motion of particles. OpenDrift is designed in a modular way, meaning it can be adapted for many different applications, including oil spill spread, drifting debris, algae movement, or search-and-rescue operations. The framework includes ready-to-use models specifically for oil drift and other floating objects, enabling scientists and environmental agencies to run simulations for both research and real-time decision-making. Overall, the paper highlights how OpenDrift simplifies trajectory modelling by providing a customizable, efficient, and data-driven platform to forecast and analyze the movement of objects in marine and atmospheric environments.

A Numerical Oil Spill Model Based on a Hybrid Method [7] presents an approach that combines two different modelling techniques to simulate the movement and spreading of oil spills on the ocean surface. The model uses both Lagrangian and Eulerian methods: the Lagrangian part tracks individual oil particles as they move with currents and wind, while the Eulerian method estimates how the oil concentration spreads and changes over a grid. By combining both methods, the model captures detailed particle movement as well as large-scale distribution of oil slicks. It considers important environmental factors such as wind, waves, ocean currents, and oil weathering processes like evaporation and dispersion. This hybrid approach improves prediction accuracy, especially in complex ocean conditions, and helps authorities understand where the oil will drift and how it will spread over time. The paper demonstrates that hybrid modelling provides a more robust and realistic simulation compared to using only one of the methods.

Influence of Sea Surface Waves on Numerical Modeling of an Oil Spill [8] examines how ocean surface waves affect the movement and behavior of oil slicks during numerical simulations. Traditional oil spill models often focus on wind and ocean currents, but this paper highlights that waves play a crucial role in determining how oil spreads on the water. Waves generate additional surface motion, known as wave-induced drift or Stokes drift, which pushes oil in different directions than currents alone would suggest. The study integrates wave data into the oil spill model and shows that including wave effects significantly improves the accuracy of trajectory predictions. When waves are ignored, the predicted oil slick position can deviate from the actual drift path, leading to incorrect planning of response operations. The paper concludes that incorporating sea surface wave dynamics into oil spill modelling results in more realistic simulations and better decision-making during spill management and containment efforts.

Table 2.1: Literature Survey Table

Paper Reference	Approach	Key Ideas	Pros	Cons
[1] Detection of oil slicks in SAR satellite images using Otsu-Bradley's thresholding method. [2] Textural Features for Image Classification.	Classical	Threshold & GCLM	Fast, Cheap	Noise Sensitive, False Alarms
[3] Oil Spill Detection using Convolutional Neural Networks and Sentinel-1 SAR Imagery. [4] Dark Spot Detection in SAR Images of Oil Spill Using Segnet. [5] U-Net: Convolutional Networks for Biomedical Image Segmentation.	DL	CNN, UNet Segmentation	High accuracy, Auto learns	Expensive, data-hungry
[6] OpenDrift v1.0: a generic framework for trajectory modelling. [7] A numerical oil spill model based on a hybrid method [8] Influence of sea surface waves on numerical modeling of an oil spill	Drift Prediction	Oil Spread Sim	Forecast future movements	Needs environment data

CHAPTER 3
PROBLEM STATEMENT

To develop a deep learning-based system that automatically detects oil slicks from SAR satellite imagery and predicts their drift, enabling faster response while reducing manual effort and misclassification due to look-alike dark regions.

CHAPTER 4
SOFTWARE REQUIREMENTS AND SPECIFICATION

4.1 PROBLEM DEFINITION

Satellite-based monitoring of marine environments often requires manual inspection to identify oil slicks and estimate their drift direction. This process is slow, prone to human error, and not suitable for real-time response in emergency situations. Radar satellite images contain complex textures and noise, making traditional image processing methods inaccurate. A system is required that automatically detects oil slicks from SAR images and predicts their drift using environmental and oceanographic data. The goal is to provide accurate oil spill detection and future movement estimation to support faster decision-making and reduce environmental damage.

4.2 FUNCTIONAL REQUIREMENTS

– SAR Image Upload : The system shall allow users to upload satellite SAR images.

– Dark Spot Detection : The system shall automatically detect and segment oil slick regions using a deep learning model (CNN)l.

– Drift Prediction : The system shall estimate future drift trajectory using numerical modeling frameworks

– Environmental Data Integration : The system shall incorporate wind, current, and wave direction data from APIs to improve drift accuracy.

– Output Visualization : The system shall display detected oil slick regions and predicted drift paths on a map interface.

4.3 NON-FUNCTIONAL REQUIREMENTS

– Performance: Oil slick detection and drift simulation must complete within an acceptable time.

– Security: Remote sensing data and API keys must be securely stored.

– Usability: The interface should be intuitive for researchers and authorities.

– Scalability: The system should handle large datasets and multiple image uploads.

– Maintainability: System must support updates to models and API integrations.

4.4 SOFTWARE AND HARDWARE REQUIREMENTS

- Software: Python (Tensorflow, PyTorch, Scikit-learn, OpenCV), Ubuntu, QGIS and GNOME.
- Hardware: Standard workstation with GPU support. (Minimum RTX4050).
- Dataset: Sentinel-1 SAR Oil spill image dataset.
- Sentinel Hub: Live SAR images for testing
- Weather Data: OpenWeatherMap API

CHAPTER 5
FLOWCHART

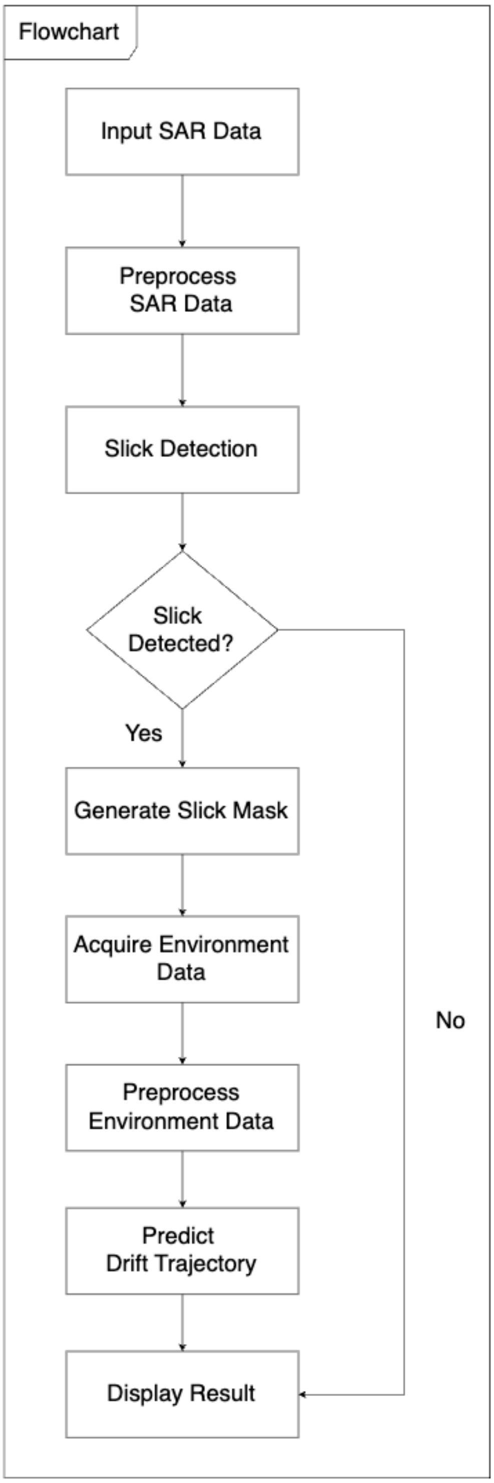

Figure 5.1: Flow Chart

CHAPTER 6
PROJECT REQUIREMENTS SPECIFICATION

6.1 METHODOLOGY

6.1.1 Data Collection

Acquire high-resolution Synthetic Aperture Radar (SAR) images from publicly available repositories such as Copernicus Open Access Hub (Sentinel-1 data). These images should cover diverse oceanic regions, seasons, and weather conditions to ensure generalization. Complement SAR imagery with environmental datasets including wind speed and direction, ocean currents, tides, and wave height sourced from NOAA or CMEMS (Copernicus Marine Environment Monitoring Service). All datasets are annotated to mark oil slick regions for supervised deep learning–based classification.

6.1.2 Image Preprocessing

Preprocess raw Sentinel-1 SAR images by performing radiometric calibration, speckle noise filtering (using Lee or Frost filters), and geometric correction to ensure spatial consistency. Normalize image intensities and crop or resample them to a uniform resolution (e.g., 256×256 pixels). Generate masks for training by manually or semi-automatically labeling oil and non-oil regions. Environmental data are cleaned and synchronized temporally with SAR acquisition timestamps.

6.1.3 Oil Slick Detection and Feature Extraction Model

Employ a Convolutional Neural Network (CNN)–based deep learning architecture to detect oil slicks. The model learns spatial and contextual features from SAR imagery to differentiate between true oil slicks and look-alike phenomena (e.g., low wind zones, natural films).

Training:

– Split the dataset into training (70%), validation (20%), and test (10%) sets.

– Optimize the network using binary cross-entropy loss with the Adam optimizer.

– Evaluate model performance using metrics like Accuracy, Precision, Recall, and F1-Score.

Validation:

– Compare predictions with labeled data using intersection-over-union (IoU) and confusion matrices.

– Fine-tune hyperparameters (learning rate, kernel size, dropout rate) for optimal accuracy.

Extract key features such as backscatter intensity, texture metrics (GLCM features), and shape descriptors to improve model discriminability. Augment the dataset using transformations such as rotation, scaling, flipping, and brightness variations to enhance robustness. Statistical descriptors of wind, tide, and current data are also encoded as auxiliary input features to assist in drift modeling.

6.1.4 Drift Prediction Model

Implement a Lagrangian particle-based model to predict the drift trajectory of detected oil slicks. The model simulates the motion of virtual oil particles influenced by wind, surface currents, wave-induced Stokes drift, and tides.

– Integrate environmental data through API calls to NOAA or CMEMS.

– Update particle positions iteratively over time to generate forecast maps.

– Optionally, incorporate a hybrid Eulerian–Lagrangian approach to combine particle track- ing with grid-based concentration modeling for improved precision.

Validation is performed by comparing simulated trajectories with observed slick positions or reference datasets.

6.1.5 System Integration and Visualization

Integrate the detection and drift prediction modules into a unified system pipeline. Develop a web-based or desktop interface using Python frameworks like Streamlit or Flask. Allow users to upload SAR images, view detected oil regions, and visualize predicted drift paths on an interactive map (e.g., via Folium or Plotly). Provide export functionality for reports and trajectory maps.

6.2 ALGORITHMS AND TECHNOLOGIES

6.2.1 Algorithms

Convolutional Neural Network (CNN):

– **Purpose:** Detect and classify oil slicks in SAR imagery.

– **Explanation:** Extracts spatial and texture features to distinguish oil slicks from look-alikes.

– **Application:** Trained on labeled Sentinel-1 data for oil spill detection.

Lagrangian Drift Model:

– **Purpose:** Simulate oil particle trajectories over time.

– **Explanation:** Models particle advection under wind, current, and wave influences.

– **Application:** Used for short-term oil drift forecasting.

Hybrid Eulerian–Lagrangian Model:

– **Purpose:** Combine concentration mapping with trajectory simulation.

– **Explanation:** Eulerian fields compute concentration, while Lagrangian particles track motion.

– **Application:** Enhances accuracy of drift prediction and dispersion assessment.

6.2.2 Technologies

– **Python:** Core programming language for model development and integration.

– **TensorFlow / PyTorch:** Frameworks for implementing deep learning models.

– **OpenCV & NumPy:** Used for image preprocessing and numerical operations.

– **Streamlit / Flask:** Frameworks for building interactive web applications.

– **NOAA / CMEMS APIs:** Provide real-time oceanographic and meteorological data for drift modeling.

– **Matplotlib & Plotly:** Used for visualization of oil slick detections and drift trajectories.

CHAPTER 7

SYSTEM ARCHITECTURE

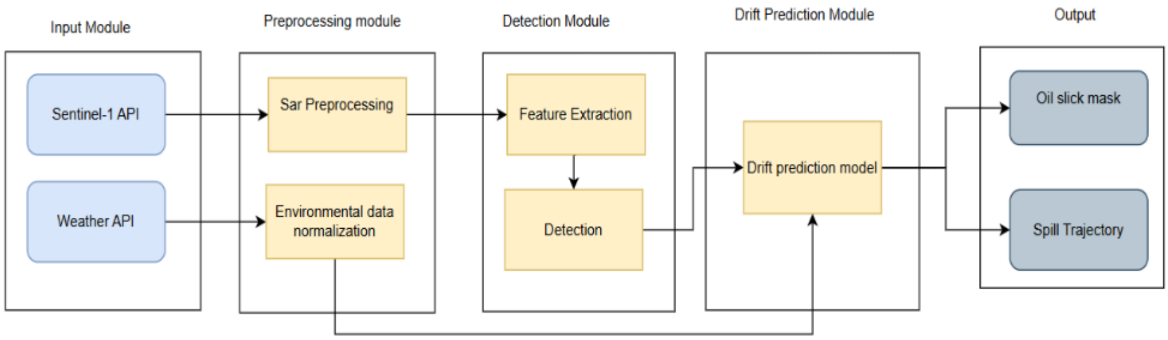

Figure 7.1: System Architecture of Oil Slick Detection and Drift Prediction

This architecture represents a complete workflow for detecting and predicting the movement of oil spills using satellite and environmental data. The pipeline begins with the Input Module, which gathers Sentinel-1 SAR imagery and weather or oceanographic data. These inputs are sent to the Preprocessing Module, where SAR images undergo essential corrections, and environmental data is cleaned and normalized for consistency.

The processed data then moves to the Detection Module, which extracts key features from SAR imagery to identify dark spots and patterns associated with oil slicks. A detection algorithm uses these features to generate an initial oil slick mask. Both the detected mask and the environmental inputs are passed to the Drift Prediction Module, where a prediction model simulates how the oil slick will move under the influence of wind, currents, and waves.

Finally, the Output Module delivers two products: an accurate oil slick mask marking spill locations, and a spill trajectory showing the forecasted drift path. This end-to-end system supports early detection, rapid response, and environmental decision-making.

CHAPTER 8
HIGH LEVEL DESIGN

8.1 DATA FLOW DIAGRAMS

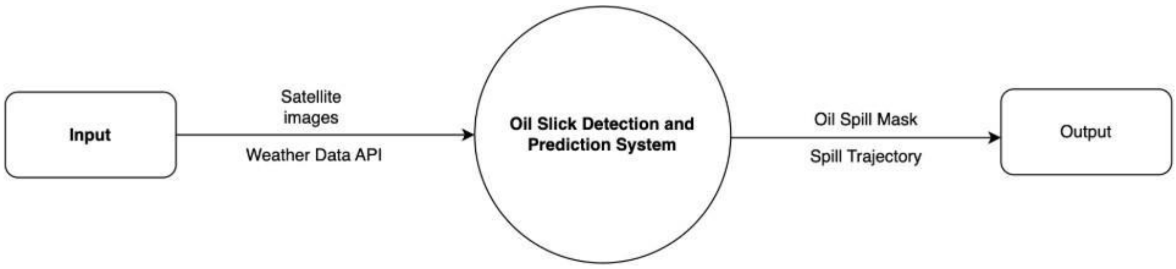

Figure 8.1: Data Flow Diagram - Level 0

The Level 0 Data Flow Diagram provides a high-level overview of the Oil Slick Detection and Prediction System. It shows how input data from satellite imagery and weather APIs enters the system and is processed to generate meaningful outputs. The system receives SAR images and environmental data, which are then analyzed by the central processing unit represented as a single process. Based on this analysis, the system produces two key outputs: an oil spill mask, which highlights detected spill regions, and a spill trajectory, which predicts the future movement of the oil slick. This top-level view simplifies the entire workflow into one main process, focusing only on the major inputs and outputs.

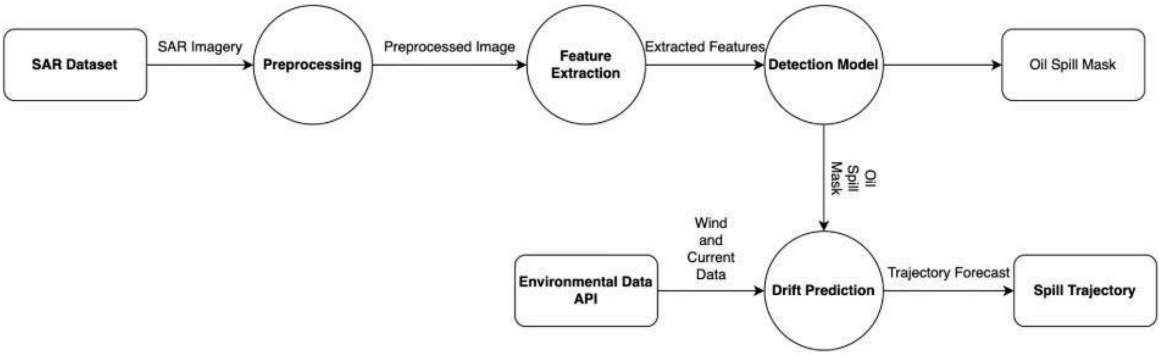

Figure 8.2: Data Flow Diagram - Level 1

The Level 1 Data Flow Diagram expands the internal processes of the system in detail. It begins with the SAR dataset, which flows into the Preprocessing module where the raw radar image is cleaned and normalized. The preprocessed image is then passed to the Feature Extraction module to identify patterns relevant to oil spills. These extracted features are fed into the Detection Model, which generates the oil spill mask. Simultaneously, environmental data such as wind and currents, obtained through the Environmental Data API, are used by the Drift Prediction module. This module combines the detected oil mask with envi- ronmental conditions to forecast the spill trajectory. The output oil spill mask and spill trajectory represent the final processed results of the system.

8.2 UML DIAGRAMS

8.2.1 Use Case Diagram

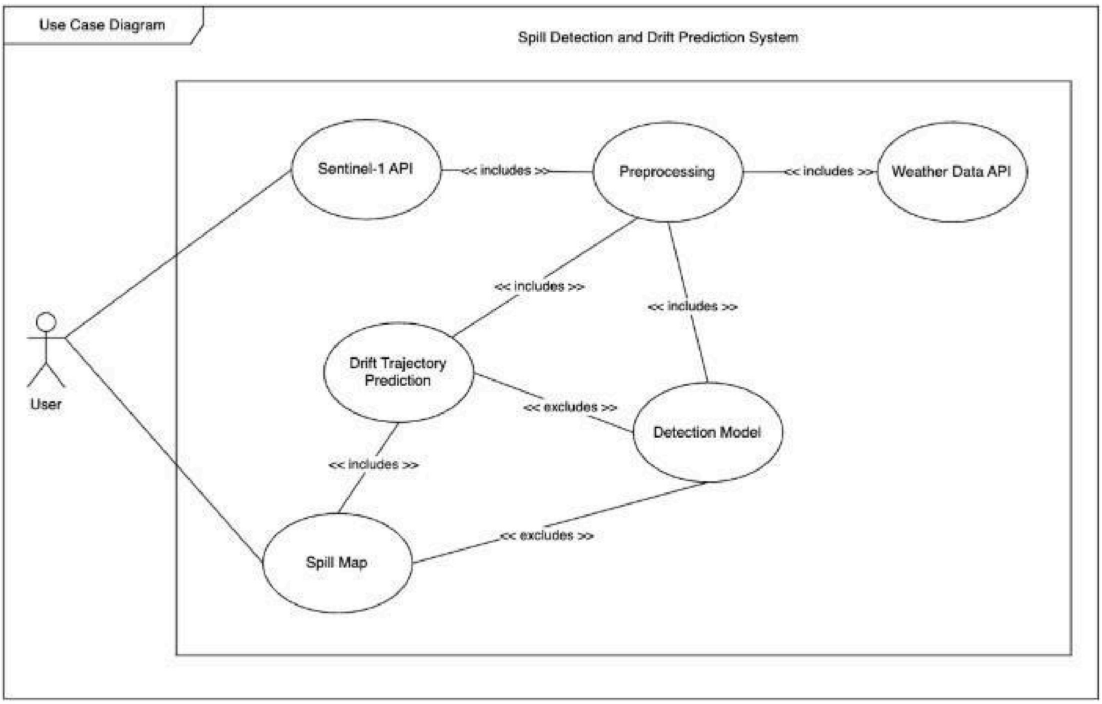

Figure 8.3: Use Case Diagram

This use case diagram explains how a user interacts with the Spill Detection and Drift Prediction System and how its core components operate together. The workflow begins when the user initiates Acquire SAR Image, which automatically triggers Preprocessing and Fetching Weather Data. These steps ensure that the SAR image is cleaned, corrected, and paired with essential environmental parameters such as wind and current data.

After preprocessing, the system may run the Detection Model, which is shown as an optional step. This model analyzes the processed SAR image to identify oil slick regions and generate a segmentation mask. The workflow then progresses to Drift Trajectory Prediction, which uses the collected environmental data to simulate how the oil slick will move over time. This prediction may operate with or without the detection model, depending on the system configuration.

Finally, the system generates a Spill Map, visually presenting the detected slick and predicted drift path. The diagram highlights how each component connects through include and exclude relationships.

8.2.2 Class Diagram

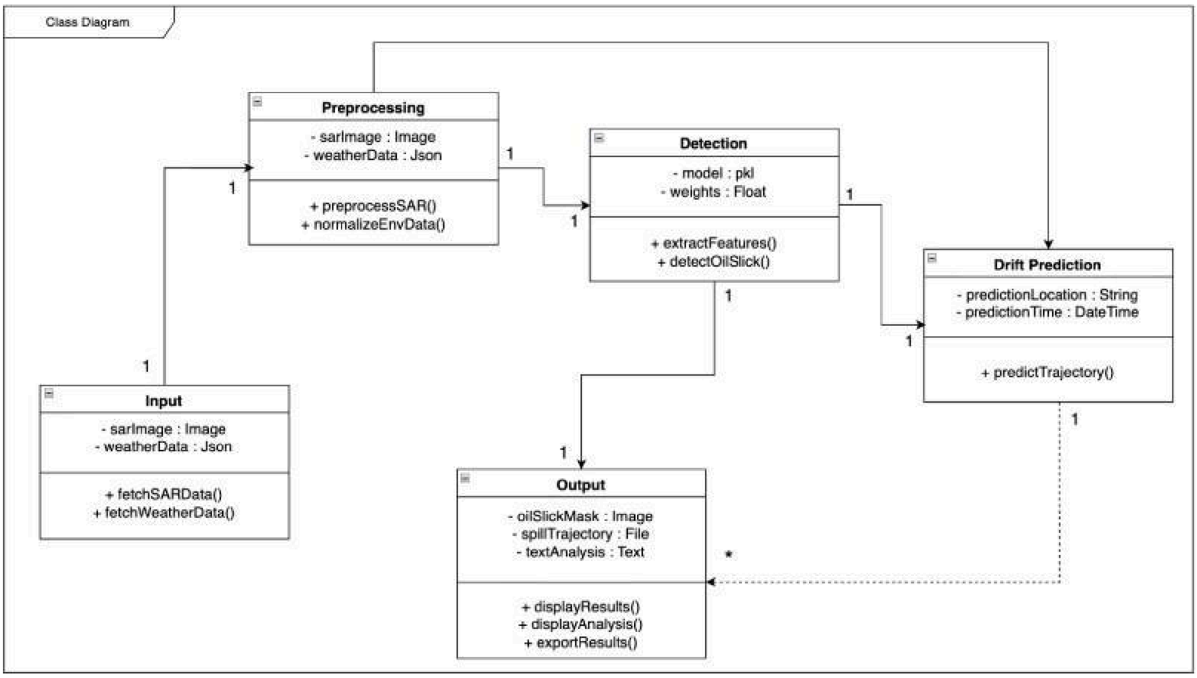

Figure 8.4: Class Diagram

This class diagram illustrates the structure and interaction between components in the Spill Detection and Drift Prediction System. The Input class collects raw SAR imagery and weather data and provides methods like fetchSARData() and fetchWeatherData() to retrieve them. This data flows into the Preprocessing class, which stores both inputs and performs essential operations such as preprocessSAR() and normalizeEnvData() to prepare the data for analysis.

The processed data is then passed to the Detection class, which contains the model and its parameters. This class handles core analytical functions including extractFeatures() and detectOilSlick(), producing a refined representation of oil slick regions. These outputs are sent to the Drift Prediction class, which uses attributes like prediction location and time to run the predictTrajectory() method and forecast how the oil slick will move.

Finally, the Output class compiles results from detection and prediction into visual and textual formats, including the oil slick mask, spill trajectory file, and summary analysis. It offers functions like displayResults() and exportResults() to present the findings to the user. The associations between classes show a clear, sequential flow where each component contributes to generating accurate and actionable output.

8.2.3 Sequence Diagram

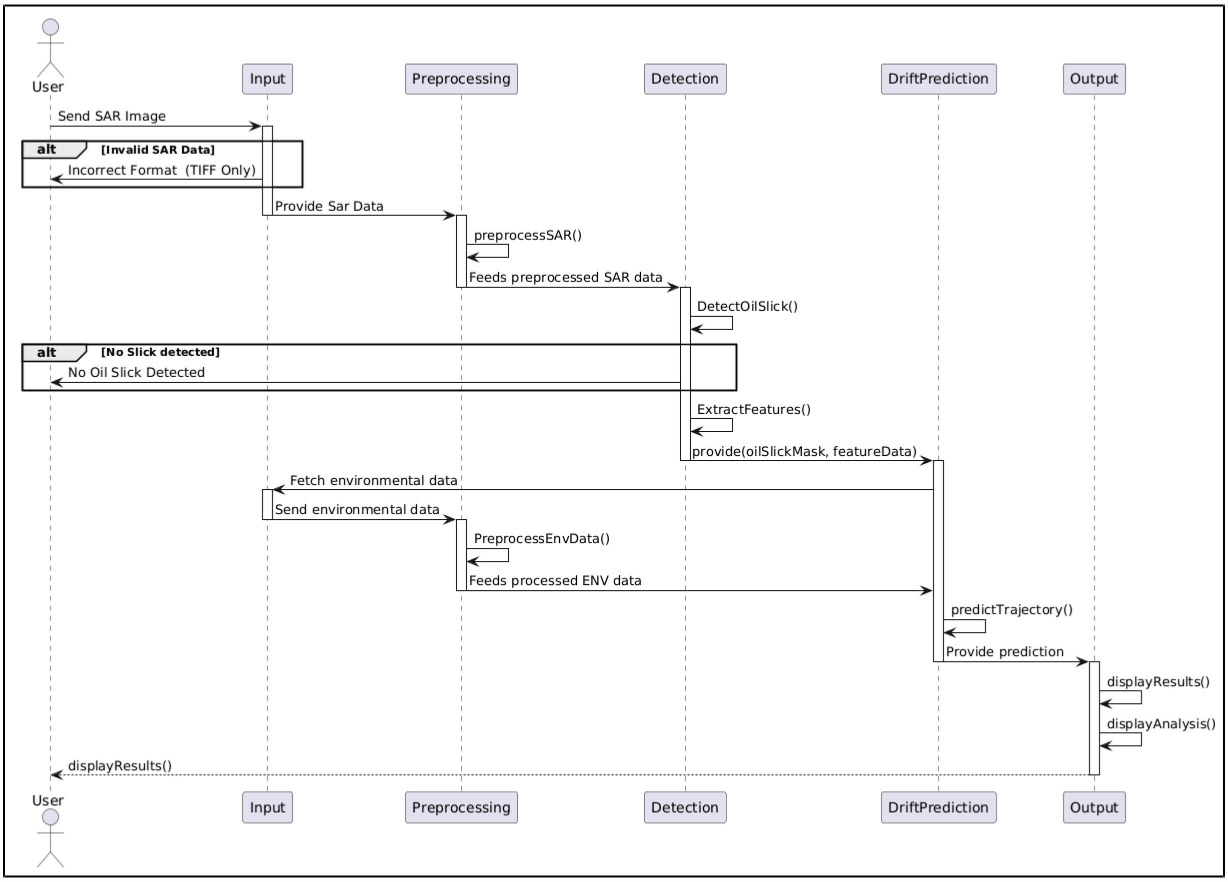

Figure 8.5: Sequence Diagram

The sequence diagram illustrates the complete workflow of an automated oil-slick detection and drift-prediction system built using SAR imagery. The interaction begins when the user provides a SAR image to the system through the Input module. The system first checks whether the uploaded file is in the correct TIFF format; if not, the user is notified about the invalid data and the process is terminated. Once the SAR data is validated, it is forwarded to the Preprocessing module.

In the preprocessing stage, the SAR image undergoes essential operations such as noise reduction, normalization, and resolution adjustment to prepare it for analysis. The preprocessed data is then sent to the Detection module, where deep-learning based algorithms attempt to identify the presence of oil slicks. If no oil slick is detected, the system directly informs the user. However, if a slick is detected, the system further extracts relevant features such as shape, intensity variation, texture, and boundary information.

These extracted features, along with the generated oil-slick mask, are passed to the DriftPrediction module. At this stage, the system fetches external environmental data, such as wind speed, ocean currents, and temperature, through the Input module. This environmental data is preprocessed and then supplied back to the drift-prediction model.

The drift-prediction model uses both the oil-slick features and environmental parameters to estimate the future movement or trajectory of the detected oil slick. A prediction output is generated and sent to the Output module.

Finally, the Output module presents the results to the user by displaying the detected slick, extracted features, predicted drift path, and relevant analytical visualizations. This ensures that users receive a complete and interpretable analysis, assisting authorities in planning response actions and understanding the progression of the oil spill.

8.2.4 Activity Diagram

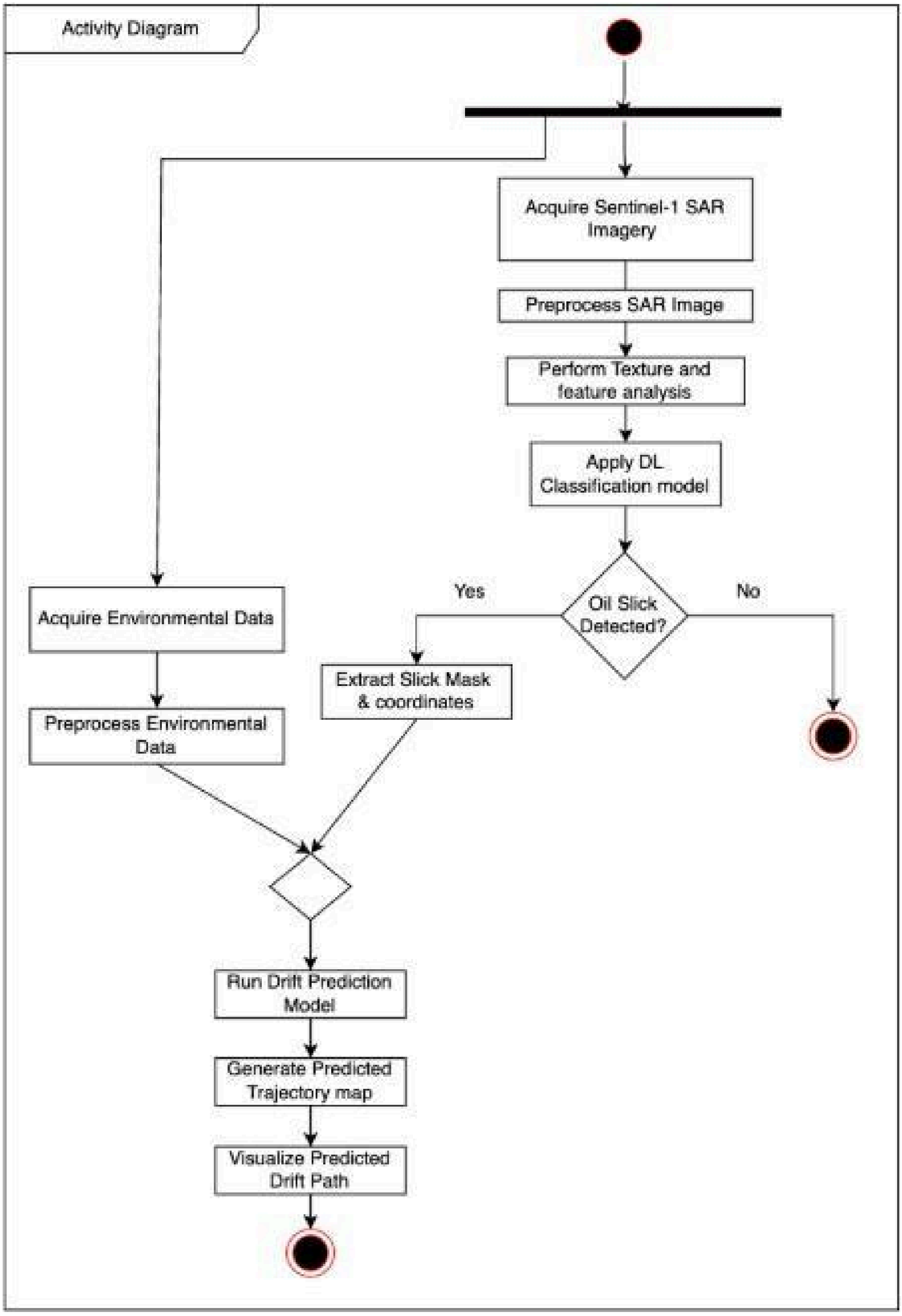

Figure 8.6: Activity Diagram

The activity diagram represents the complete workflow of the oil-slick detection and drift-prediction system, outlining how data flows from raw satellite imagery to the final predicted drift path. The process begins with the acquisition of Sentinel-1 SAR imagery, which serves as the primary input for the detection pipeline. Once the SAR image is obtained, it is preprocessed to remove noise, normalize intensity values, and enhance overall image quality.

Following preprocessing, the system performs texture extraction and feature analysis to highlight characteristics that may indicate the presence of an oil slick. These features are then fed into a deep learning–based classification model, which determines whether an oil slick is present. If no slick is detected, the process terminates, and no further analysis is conducted. If a slick is detected, the system extracts the slick mask and its geographical coordinates for further processing.

Simultaneously, the workflow also gathers environmental data such as wind fields, ocean currents, and sea-surface conditions, which are essential for drift modelling. This environmental data undergoes preprocessing to ensure compatibility and consistency with the drift-prediction system. Once both the oil-slick features and environmental parameters are prepared, they are merged and passed into the drift-prediction model.

The drift-prediction model simulates the movement of the slick over time, considering dynamic environmental factors that influence its trajectory. The system then generates a predicted trajectory map showing how the slick is expected to move in the near future. Finally, this predicted drift path is visualized and presented as the output, enabling responders and analysts to make informed decisions for mitigation and cleanup planning.

CHAPTER 9
SYSTEM IMPLEMENTATION

9.1 TECHNOLOGY STACK

9.1.1 Programming Languages

– **Python** – Used for data preprocessing, SAR image analysis, deep learning model training, drift simulation, and backend workflow automation.

9.1.2 Frameworks and Libraries Deep Learning

– **TensorFlow / Keras** – For building CNN, U-Net, and SegNet-based oil spill detection models.

– **PyTorch** – Used for advanced segmentation model development and experimentation.

– **Scikit-Learn** – For feature engineering and evaluation metrics.

– **OpenCV** – Noise reduction, filtering, and preprocessing of SAR images.

– **NumPy** – Numerical operations and matrix computations.

– **Rasterio** – Handling geospatial raster data from SAR satellites.

9.1.3 Satellite and Environmental Data Sources

– **Sentinel-1 SAR Dataset** – Primary dataset for oil slick detection.

– **Sentinel Hub** - Live testing Images

– **OpenWeatherMap API** – Provides wind, current, temperature, and environmental parameters based on the latitude and longitude provided.

9.1.4 Drift Prediction Models and Tools

– **OpenDrift Framework** – Lagrangian particle-based drift simulation.

9.1.5 GIS and Visualization Tools

- **QGIS** – Geospatial visualization of detected masks and drift trajectories.
- **Matplotlib / Seaborn / Plotly** – Visualization of training results and forecast graphs.

9.1.6 Data Storage and File Formats

- **JSON** – Storage format for weather API data.
- **GeoTIFF / PNG** – For SAR images and segmentation mask outputs.
- **CSV** – For trajectory logs, numerical results, and dataset metadata.

9.2 INPUT MODULE

Overview

The Input Module handles the acquisition of all external data required by the system. This includes retrieving SAR imagery from satellite sources and collecting environmental data via APIs. It ensures proper formatting, validation, and consistency of incoming data.

Workflow

- Acquires Sentinel-1 SAR images using APIs or dataset repositories.
- Requests environmental parameters such as wind, temperature, and ocean current data from the Weather API.
- Validates whether the files are of required format.
- Combines both the (red and green) bands into single image.
- Forwards all input data to the Preprocessing Module.

9.3 PREPROCESSING MODULE

Overview

The Preprocessing Module is responsible for enhancing and standardizing the raw SAR imagery and environmental datasets before analysis. Since Sentinel-1 SAR images often contain speckle noise, radiometric distortions, and varying resolutions, preprocessing ensures that all data is clean, uniform, and suitable for accurate detection and prediction.

Workflow

- Receives raw SAR imagery from the Sentinel-1 API or dataset repository.
- Applies speckle noise reduction using filters such as Lee or median filtering.
- Performs radiometric calibration and geometric corrections on all SAR images.
- Normalizes and resizes imagery to maintain consistent input dimensions for deep learning models.
- Outputs enhanced, standardized SAR images to the Detection Module.

9.4 DETECTION MODULE

Overview

The Detection Module identifies and segments oil slick regions from preprocessed SAR images using deep learning techniques. Models such as CNN, U-Net, and SegNet analyze spatial and textural features to distinguish oil slicks from look-alikes (such as low-wind areas or algae) and clean water.

Workflow

- Accepts preprocessed SAR images from the Preprocessing Module.
- Extracts spatial and contextual features using deep learning architectures.
- Classifies region into oil slick, look-alike, or clean water categories.
- Generates a segmentation mask highlighting detected slick boundaries.

- Passes the oil slick mask to the Drift Prediction Module for further simulation.

9.5 DRIFT PREDICTION MODULE

Overview

The Drift Prediction Module forecasts the future movement of the detected oil slick using environmental parameters. It integrates wind speed, current direction, wave height, and temperature data to simulate oil displacement using Lagrangian, Eulerian, or hybrid drift modeling techniques.

Workflow

- Receives the detected oil slick mask from the Detection Module.
- Fetches environmental data such as wind and surface currents from the Weather API.
- Simulates oil slick transport using drift models like OpenDrift or custom hybrid numerical models.
- Predicts the future trajectory and spread of the oil slick.
- Outputs a trajectory map for visualization and further analysis.

9.6 OUTPUT AND VISUALIZATION MODULE

Overview

The Output Module compiles and visualizes the final results produced by the system. It presents both the detected oil spill mask and the predicted drift trajectory in readable and exportable formats, suitable for monitoring, reporting, or decision-making.

Workflow

- Receives the oil slick mask from the Detection Module.
- Receives trajectory forecasts from the Drift Prediction Module.
- Visualizes all results using mapping tools or plotting libraries.
- Displays consolidated results to the user for interpretation.

CHAPTER 10

CONCLUSIONS AND FUTURE WORK

Oil slick detection and drift prediction are vital for safeguarding marine ecosystems, guiding emergency response, and enforcing maritime regulations. Over the years, methods have evolved significantly from conventional interpretation and manual thresholding to advanced machine learning and deep learning techniques and frameworks. Otsu and Bradley are computationally efficient, but they often fail in complex sea states due to look-alike features. Machine learning and deep learning methods techniques have improved automation but they still heavily depend on handcrafted features or labeled datasets. Deep Learning has revolutionized the field, with CNNs, U-Net, and SegNet models achieving higher accuracy and by learning directly from large-scale SAR datasets.

Similarly, drift prediction has progressed from conventional methods like Lagrangian and Eulerian models to hybrid, ensemble, and data assimilation in order to capture better nonlinear ocean dynamics. The use of SAR images, optical, and environment data has increased the reliability, enhancing real-time monitoring and forecasting of oil slick.

Despite such advances in the technology some challenges arise and remain such as limited availability of datasets, high computational power and lack of standardized metrics. Future studies and research focus on hybrid detection, physics-informed AI, and global benchmark datasets. Addressing these gaps, next-generation systems can deliver scalable, real-time, and globally deployable solutions for marine oil spill monitoring and management.

CHAPTER 11
BIBLIOGRAPHY

[1] C. Popa, D. Atodiresei, A. Toma, V. Dobref, and J. Vatamanu, "Solutions for Modelling the Marine Oil Spill Drift," *Environments*, vol. 12, no. 4, p. 132, Apr. 2025, doi: 10.3390/environments12040132.

[2] C. Dearden, T. Culmer, and R. Brooke, "Performance Measures for Validation of Oil Spill Dispersion Models Based on Satellite and Coastal Data," *IEEE Journal of Oceanic Engineering*, vol. 47, no. 1, pp. 126–140, Sep. 2021, doi: 10.1109/joe.2021.3099562.

[3] P. Berens, "Introduction to Synthetic Aperture Radar (SAR)," 2006. Available: apps.dtic.mil/sti/tr/pdf/ADA470686.pdf.

[4] F. Mahdikhani and M. Hassannejad Bibalan, "Detection of Oil Slicks in SAR Satellite Images Using Otsu-Bradley's Thresholding Method," *Majlesi Journal of Electrical Engineering*, vol. 19, no. 2, Jun. 2025, doi: 10.57647/j.mjee.2025.1902.24.

[5] J. Xu *et al.*, "Oil Slick Identification in Marine Radar Image Using HOG, Random Forest and PSO," *IEEE Geoscience and Remote Sensing Letters*, vol. 21, pp. 1–5, Jan. 2024, doi: 10.1109/lgrs.2024.3431043.

[6] E. Kalogirou *et al.*, "Oil Spill Detection Using Convolutional Neural Networks and Sentinel-1 SAR Imagery," *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. XLVIII-G-2025, pp. 757–764, Jul. 2025, doi: 10.5194/isprs-archives-xlviii-g-2025-757-2025.

[7] H. Guo, G. Wei, and J. An, "Dark Spot Detection in SAR Images of Oil Spill Using SegNet," *Applied Sciences*, vol. 8, no. 12, p. 2670, Dec. 2018, doi: 10.3390/app8122670.

[8] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional Networks for Biomedical Image Segmentation," 2015. Available: arxiv.org/pdf/1505.04597.pdf.

[9] D. Xiang, Y. Lu, D. Guan, G. Li, J. Cheng, and B. Li, "Oil Spill Detection in PolSAR Imagery Using Composite Scattering Power Entropy and Multi-Scale Hybrid Feature Fusion Network," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, pp. 1–20, Jan. 2025, doi: 10.1109/jstars.2025.3570697.

[10] R. M. Haralick, K. Shanmugam, and I. Dinstein, "Textural Features for Image Classifi- cation," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. SMC-3, no. 6, pp. 610–621, Nov. 1973, doi: 10.1109/tsmc.1973.4309314.

[11] A. S. Mahmoud, S. A. Mohamed, R. A. El-Khoriby, H. M. AbdelSalam, and I. A. El-Khodary, "Oil Spill Identification Based on Dual Attention U-Net Model Using Synthetic Aperture Radar Images," *Journal of the Indian Society of Remote Sensing*, vol. 51, no. 1, pp. 121–133, Nov. 2022, doi: 10.1007/s12524-022-01624-6.

[12] K.-F. Dagestad, J. Röhrs, Ø. Breivik, and B. Ådlandsvik, "OpenDrift v1.0: A Generic Framework for Trajectory Modelling," *Geoscientific Model Development*, vol. 11, no. 4, pp. 1405–1420, Apr. 2018, doi: 10.5194/gmd-11-1405-2018.

[13] M. De Dominicis, N. Pinardi, G. Zodiatis, and R. Lardner, "MEDSLIK-II: A Lagrangian Marine Surface Oil Spill Model for Short-Term Forecasting – Part 1: Theory," *Geoscien- tific Model Development*, vol. 6, no. 6, pp. 1851–1869, Nov. 2013, doi: 10.5194/gmd-6-
1851-2013.

[14] D. Liu, Y. Li, and L. Mu, "Parameterization Modeling for Wind Drift Factor in Oil Spill Drift Trajectory Simulation Based on Machine Learning," *Frontiers in Marine Science*, vol. 10, Jul. 2023, doi: 10.3389/fmars.2023.1222347.

[15] B. G. Gautama, N. Longepe, R. Fablet, and G. Mercier, "Assimilative 2-D Lagrangian Transport Model for the Estimation of Oil Leakage Parameters from SAR Images: Application to the Montara Oil Spill," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 9, no. 11, pp. 4962–4969, Nov. 2016, doi: 10.1109/jstars.2016.2606110.

[16] W. J. Guo and Y. X. Wang, "A Numerical Oil Spill Model Based on a Hybrid Method," *Marine Pollution Bulletin*, vol. 58, no. 5, pp. 726–734, May 2009, doi: 10.1016/j.marpolbul.2008.12.015.

[17] W. Shao *et al.*, "Influence of Sea Surface Waves on Numerical Modeling of an Oil Spill: Revisit of Symphony Wheel Accident," *Journal of Sea Research*, vol. 201, pp. 102529–102529, Aug. 2024, doi: 10.1016/j.seares.2024.102529.

[18] K. Kampouris, V. Vervatis, J. Karagiorgos, and S. Sofianos, "Oil Spill Model Uncertainty Quantification Using an Atmospheric Ensemble," *Ocean Science*, vol. 17, no. 4, pp. 919–934, Jul. 2021, doi: 10.5194/os-17-919-2021.

[19] P. Keramea *et al.*, "Satellite Imagery in Evaluating Oil Spill Modelling Scenarios for the Syrian Oil Spill Crisis, Summer 2021," *Frontiers in Marine Science*, vol. 10, Oct. 2023, doi: 10.3389/fmars.2023.1264261.

[20] S. Fan, L.-Y. Oey, and P. Hamilton, "Assimilation of Drifter and Satellite Data in a Model of the Northeastern Gulf of Mexico," *Continental Shelf Research*, vol. 24, no. 9, pp.

1001–1013, May 2004, doi: 10.1016/j.csr.2004.02.013.

[21] F. Yu, W. Sun, J. Li, Y. Zhao, Y. Zhang, and G. Chen, “An Improved Otsu Method for Oil Spill Detection from SAR Images,” *Oceanologia et Hydrobiologia*, vol. 59, no. 3, pp. 311–317, Jul. 2017, doi: 10.1016/j.oceano.2017.03.005.

[22] K. Li, H. Yu, Y. Xu, and X. Luo, “Detection of Oil Spills Based on Gray Level Co-occurrence Matrix and Support Vector Machine,” *Frontiers in Environmental Science*, vol. 10, Dec. 2022, doi: 10.3389/fenvs.2022.1049880.

[23] M. Reinan *et al.*, “SAR Oil Spill Detection System Through Random Forest Classifiers,” *Remote Sensing*, vol. 13, no. 11, p. 2044, May 2021, doi: 10.3390/rs13112044.

[24] L. Lopez, M. Moctezuma, and F. Parmiggiani, “Oil Spill Detection Using GLCM and MRF,” Nov. 2005, doi: 10.1109/igarss.2005.1526349.